

Complex Numbers

- **Exercise 1**

Compute

$$1 + i + i^2 + \dots + i^{2026}.$$

- **Solution 1.1**

We have that

$$i^n = \begin{cases} 1 & (n = 4k) \\ i & (n = 4k + 1) \\ -1 & (n = 4k + 2) \\ -i & (n = 4k + 3) \end{cases} \quad \text{For some } k \in \mathbb{Z}.$$

Thus

$$1 + i + i^2 + \dots + i^{2026} = 1 + i - 1 - i + \dots + 1 + i - 1 = i.$$

- **Exercise 2**

Prove that $\forall z \in \mathbb{C}, z = \bar{z} \iff z$ is real, $z = -\bar{z} \iff z$ is purely imaginary.

- **Proof 2.1**

Suppose $x, y \in \mathbb{R}, z = x + iy$. Then $\bar{z} = x - iy$. Suppose $z = \bar{z}$.

1. $(\implies) x + iy = x - iy \implies x = x, y = y \implies y = 0 \implies z = x \in \mathbb{R}.$

$(\impliedby)$ If $z \in \mathbb{R}$ then $z = z + 0i$ and then $\bar{z} = z - 0i = z$.

2. $(\implies)$ Suppose $z = -\bar{z}$, then $x + iy = -x + iy \implies x = -x \implies x = 0, z = 0 + iy = iy$ is a purely imaginary.

$(\impliedby)$ Suppose z is a purely imaginary, $z = 0 + iy, y \in \mathbb{R}, \bar{z} = 0 - iy, -\bar{z} = iy \implies z = -\bar{z}.$

- **Exercise 3**

Describe the sets of all $z \in \mathbb{C}$ satisfying:

1. $|z| = 2.$

2. $|z - (1 + i)| = 2.$

3. $|z - (2 - i)| < 3.$

- **Solution 3.1**

Let $x, y \in \mathbb{R}$ and let $z = x + iy = (x, y)$ represent a point in the complex plane. Geometrically, for any fixed $z_0 \in \mathbb{C}$, the expression $|z - z_0|$ denotes the distance between the points z and z_0 . We analyze each set as follows:

1. **Description of the first set ($|z| = 2$):**

By the definition of the modulus, substituting $z = x + iy$ yields:

$$|x + iy| = 2 \implies \sqrt{x^2 + y^2} = 2.$$

Squaring both sides, we obtain the algebraic equation:

$$x^2 + y^2 = 4.$$

Geometrically, this equation represents a circle in the complex plane.

Therefore, the set describes a **circle centered at the origin $(0, 0)$ with a radius of 2.**

2. **Description of the second set ($|z - (1 + i)| = 2$):**

We rewrite the equation by grouping the real and imaginary parts:

$$|(x - 1) + i(y - 1)| = 2.$$

By the definition of the modulus, squaring both sides yields:

$$(x - 1)^2 + (y - 1)^2 = 4.$$

Geometrically, this is the standard Cartesian equation of a circle.

Therefore, the set describes a **circle centered at the point $1 + i = (1, 1)$ with a radius of 2**.

3. **Description of the third set ($|z - (2 - i)| < 3$):**

We rewrite the inequality by grouping the real and imaginary parts:

$$|(x - 2) + i(y + 1)| < 3.$$

By the definition of the modulus, squaring both sides yields:

$$(x - 2)^2 + (y + 1)^2 < 9.$$

Geometrically, a strict inequality ($<$) represents the interior region bounded by a boundary curve.

Therefore, the set describes the **open disk consisting of all points inside the circle centered at $2 - i = (2, -1)$ with a radius of 3 (excluding the boundary circle itself)**.

• **Exercise 4**

Describe the set of all complex numbers z satisfying $|z - 1| = |z - i|$.

◦ **Solution 4.1**

Let $x, y \in \mathbb{R}$ and let $z = x + iy$. We substitute z into the modulus equation:

$$|(x - 1) + iy| = |x + i(y - 1)|.$$

By the definition of the modulus, squaring both sides yields:

$$(x - 1)^2 + y^2 = x^2 + (y - 1)^2.$$

Expanding both sides of the equation, we obtain:

$$x^2 - 2x + 1 + y^2 = x^2 + y^2 - 2y + 1.$$

Canceling out the common terms x^2 , y^2 , and 1 from both sides gives:

$$-2x = -2y \implies y = x.$$

Geometrically, this equation represents a straight line in the complex plane.

Therefore, the set describes the straight line $y = x$, which is the perpendicular bisector of the line segment joining the points $1 = (1, 0)$ and $i = (0, 1)$.

• **Exercise 5**

Without using polar form, find all complex numbers satisfying $z^2 = i$.

◦ **Solution 5.1**

Let $x, y \in \mathbb{R}$ and let $z = x + yi$. Then squaring z yields $z^2 = x^2 - y^2 + 2xyi = 0 + 1i$. By equalling the real and imaginary parts, we obtain the following system of equations:

$$x^2 - y^2 = 0, \quad (1)$$

$$2xy = 1. \quad (2)$$

From equation (1), factoring the difference of squares gives $(x - y)(x + y) = 0$, which implies $x = y$ or $x = -y$.

From equation (2), we have $xy = \frac{1}{2} > 0$, which means x and y must have the same sign. This rules out the case $x = -y$. Therefore, we must have $x = y$.

Substituting $x = y$ into equation (2) yields $2x^2 = 1 \implies x^2 = \frac{1}{2}$, which gives $x = \pm \frac{\sqrt{2}}{2}$. We then split it into two cases based on the sign:

1. **Case 1:** $x, y > 0$.

This yields $x = y = \frac{\sqrt{2}}{2}$. Hence, $z = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$.

2. **Case 2:** $x, y < 0$.

This yields $x = y = -\frac{\sqrt{2}}{2}$. Hence, $z = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$.

Therefore, the required complex numbers satisfying $z^2 = i$ are $z = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$ and $z = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$.